

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 6 · Regression modeling continued

TUESDAY SEP 29 AND THURSDAY OCT 1, 2026

WHAT YOU WILL BE ABLE TO DO BY MONDAY NIGHT

- 1 Decide when a question calls for a straight line, a curve, a dummy variable, or an interaction term.
- 2 Use residuals to create easily interpreted adjusted player metrics.



UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials.



3 Confirm that today's deck is in spax402-course-materials/decks/.

Thursday's files arrive through /update before Thursday's class.

01

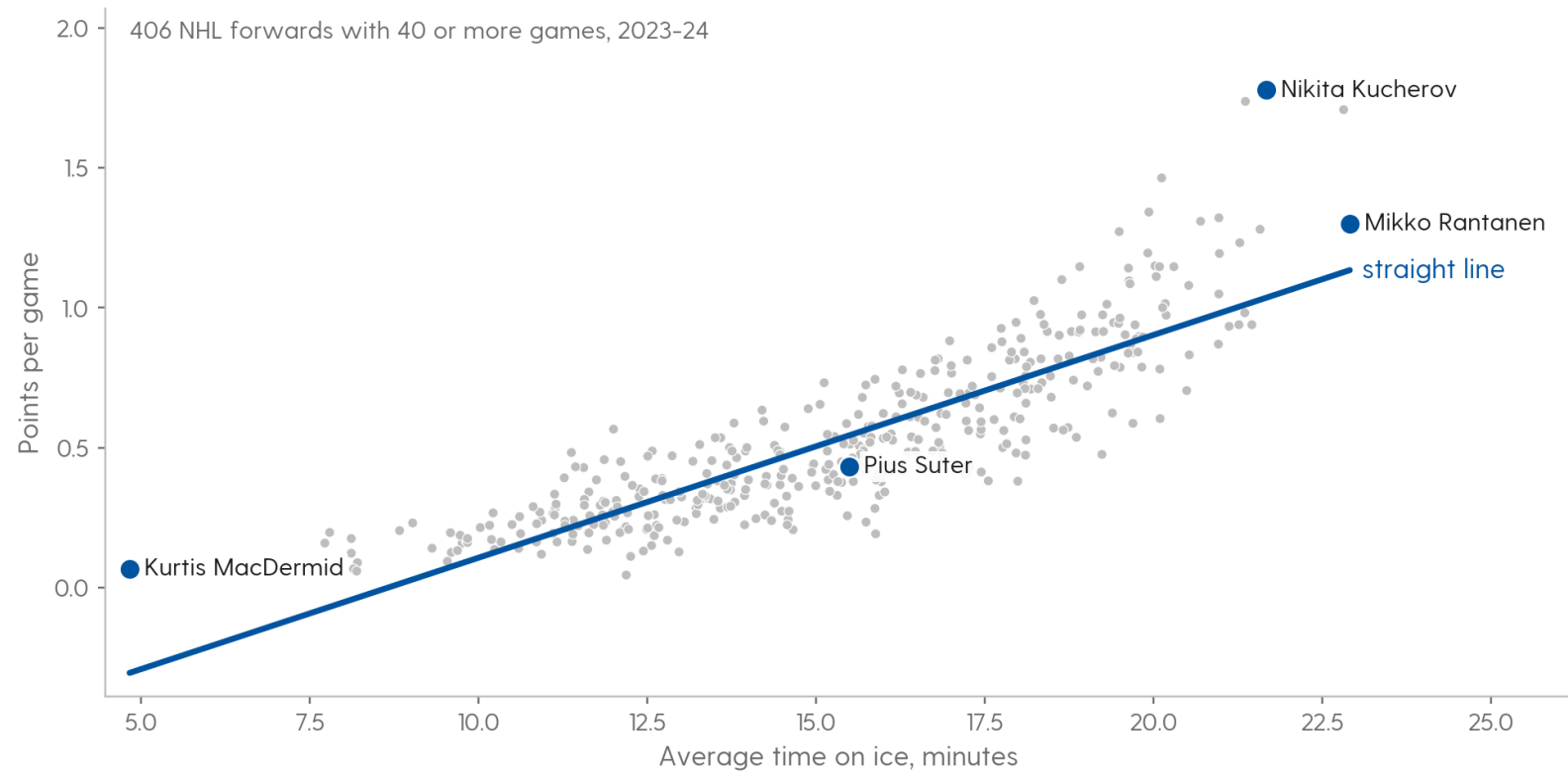
NON-LINEAR RELATIONSHIPS



DO PLAYERS SCORE MORE OR FEWER POINTS PER MINUTE IF THEY PLAY MORE MINUTES PER GAME?

CURVES

POINTS PER GAME AND TIME ON ICE, 2023-24



R squared 0.74 across 406 forwards.

NHL stats API, skater summary, regular season 2023-24. Forwards with 40 or more games.

NON-LINEAR RELATIONSHIP

As the independent variable increases, the rate of change of the dependent variable itself changes. It may increase, decrease, or even change direction.

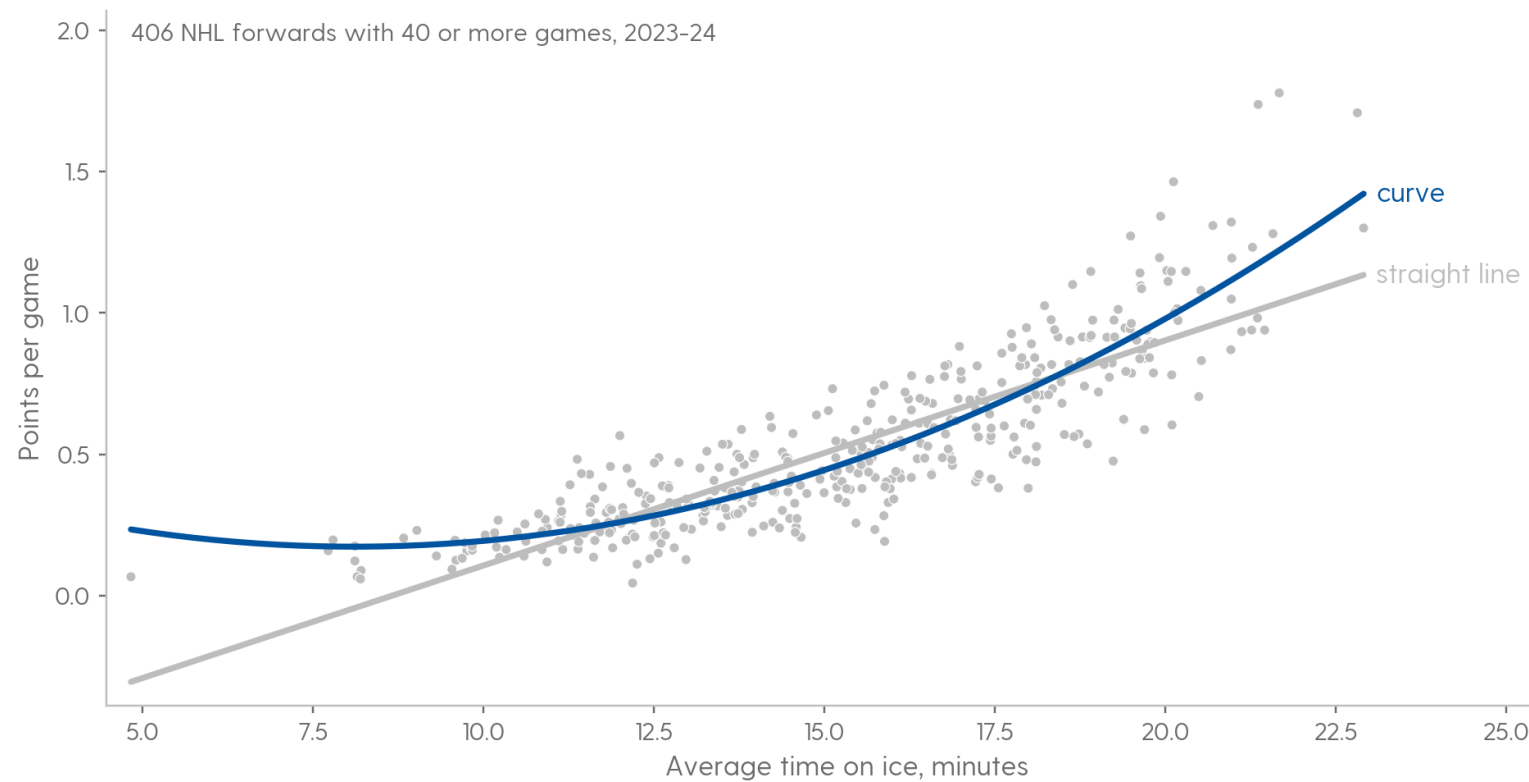
QUADRATIC REGRESSION

To create a quadratic regression (Polynomial regression with degree of two), we'll just need to create an additional independent variable, x^2 . We can then create a model using our original and calculated independent variables.

$y = a + b_1 \times x + b_2 \times x^2$. In Excel, add a column that squares x and give LINEST both columns: LINEST(y range, x and x^2 columns, TRUE, TRUE).

CURVES

THE STRAIGHT LINE AND THE CURVE, 2023-24



R squared rises from 0.74 to 0.79. The curve's x^2 coefficient is 0.0057.

NHL stats API, skater summary, regular season 2023-24. Forwards with 40 or more games.

SUTER AND KUCHEROV, STRAIGHT LINE AND CURVE

Straight line: $PPG = -0.690 + 0.0796 \times TOI$

Curve: $PPG = 0.547 - 0.0923 \times TOI + 0.00570 \times TOI^2$

Pius Suter, 15.5 minutes: line 0.54, curve 0.48, actual 0.43

Nikita Kucherov, 21.7 minutes: line 1.04, curve 1.22, actual 1.78

The squared term adds 1.37 points per game to Suter's prediction and 2.67 to Kucherov's. NHL stats API, skater summary, regular season 2023–24. Forwards with 40 or more games.

SEVERINI'S CURVE FOR NHL FORWARDS

$$\text{points per game} = 0.0905 - 0.0201 \times \text{TOI} + 0.00280 \times \text{TOI}^2$$

As time on ice increases, the marginal gain in points per additional minute also increases.

TOI is average time on ice in minutes. Severini (2020), Analytic Methods in Sports.

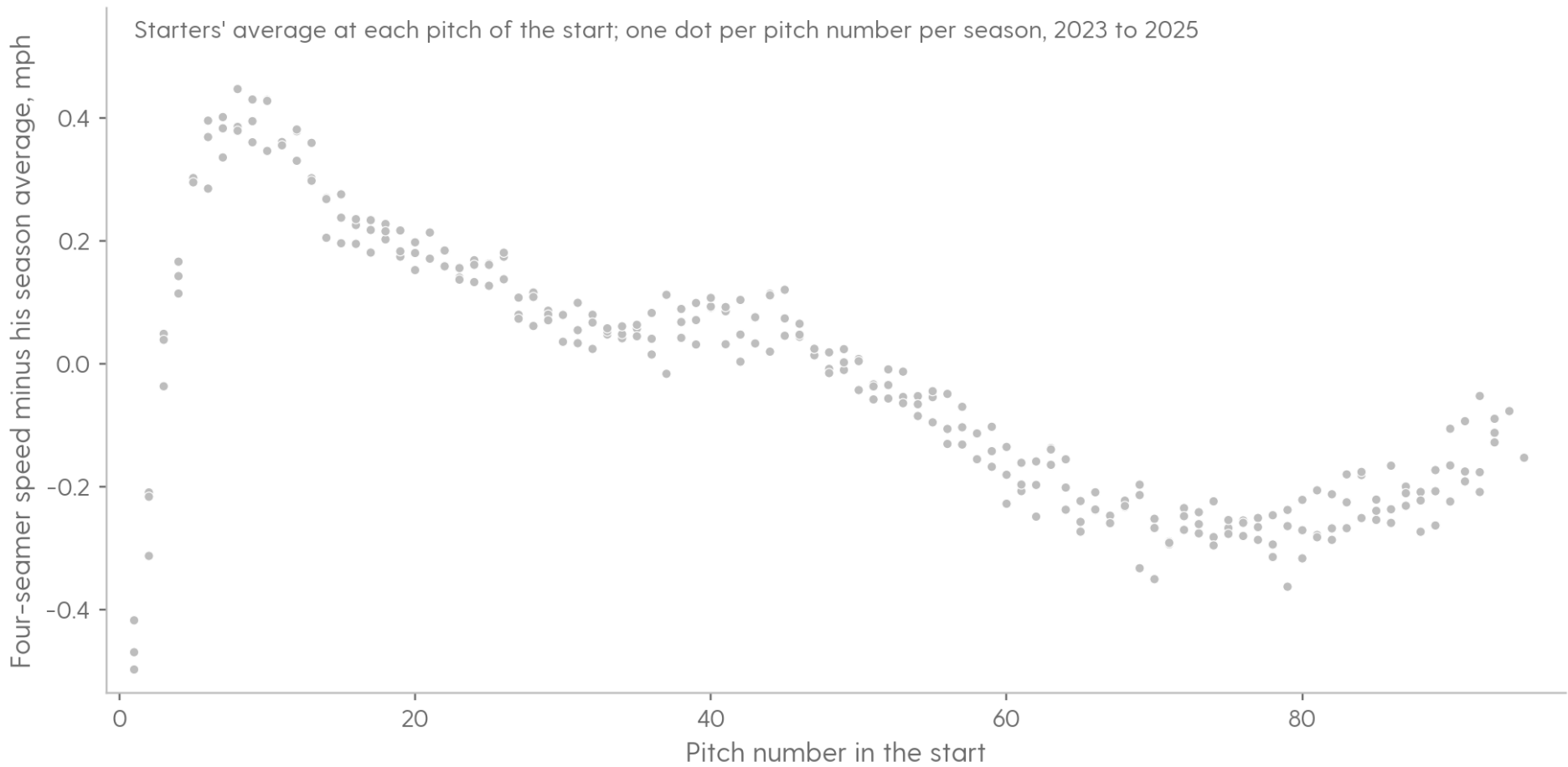
CURVES



WHY?

? HOW MANY TIMES WOULD YOU EXPECT A STARTER'S FASTBALL VELOCITY TO TURN OVER A START?

FASTBALL VELOCITY BY PITCH OF THE START, 2023 TO 2025



Baseball Savant Statcast search, 2023 to 2025 regular seasons. Four-seam fastballs by starting pitchers, every pitch type counted in the pitch number, pitch counts with 500 or more fastballs.

CURVES

FIT ON 2023 AND 2024, CHECKED ON 2025

DEGREE	TURNS INSIDE THE DATA	R SQUARED, 2023 TO 2024	R SQUARED, 2025
1	0	0.014	0.017
2	0	0.014	0.017
3	2	0.021	0.025
4	2	0.022	0.026
5	2	0.026	0.029
6	5	0.029	0.033
7	4	0.031	0.034

A degree-n polynomial can turn at most $n - 1$ times; the fit decides whether it uses them.

Baseball Savant Statcast search, 2023 to 2025 regular seasons. Four-seam fastballs by starting pitchers, every pitch type counted in the pitch number, pitch counts with 500 or more fastballs.

02

HEIGHT, WEIGHT AND THE 40

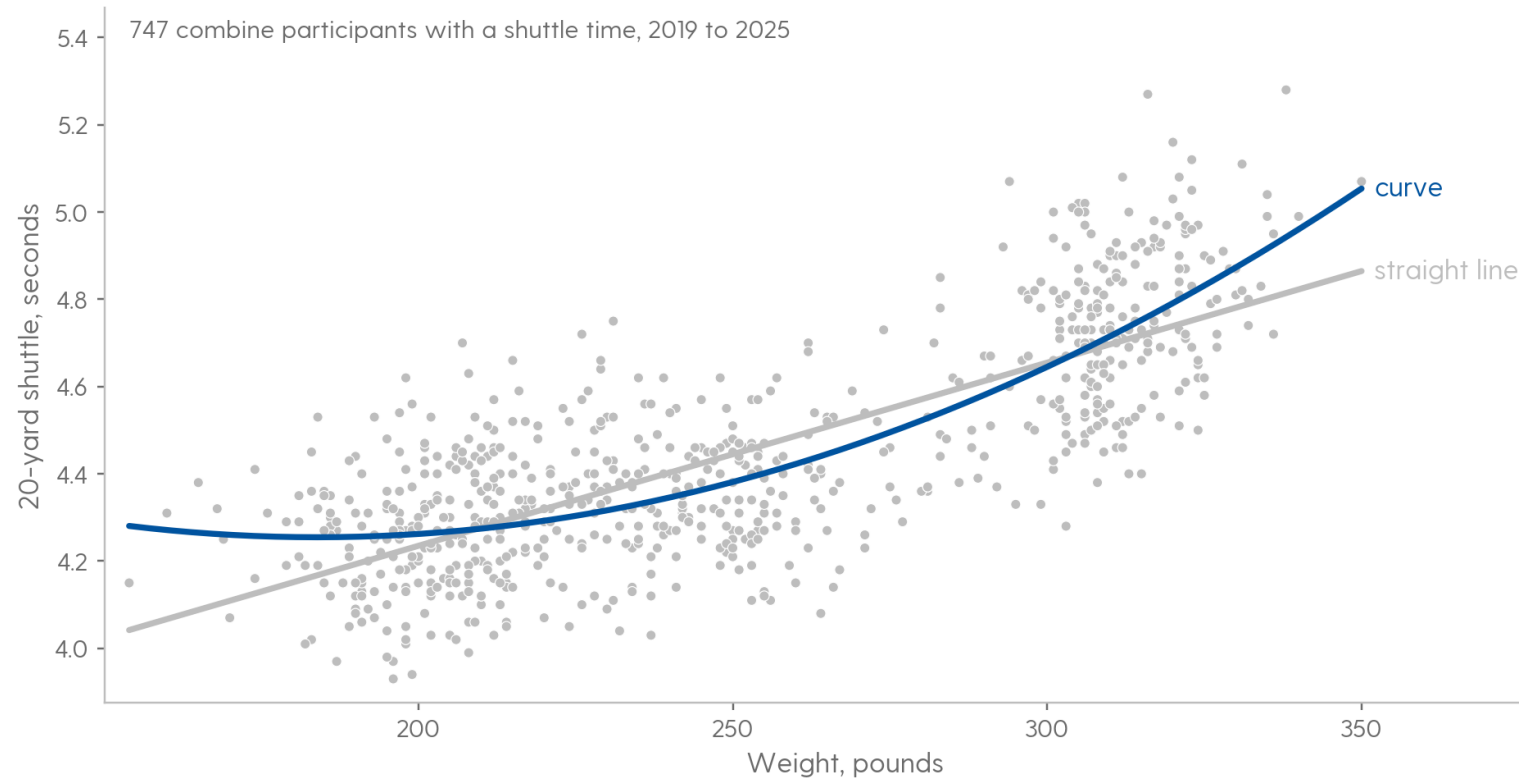
HEIGHT AND WEIGHT: LINEAR OR NON-LINEAR RELATIONSHIP?



HOW MANY DIMENSIONS DO WE GROW IN?

WEIGHT AND 40-YARD DASH TIME: LINEAR OR NON-LINEAR RELATIONSHIP?

WEIGHT AND SHUTTLE TIME, 2019 TO 2025 COMBINES



R squared 0.60 with a straight line and 0.64 with a curve, and the weight^2 term has $p < 0.0001$.

nflverse combine release file, 2019 to 2025 combines, every participant with a weight and a shuttle time. 2021 is left out: with no in-person combine, its times are hand-timed pro days.



XAVIER WORTHY RAN A 4.21 AT 165 POUNDS IN 2024. JORDAN DAVIS RAN A 4.78 AT 341 POUNDS IN 2022. WHO IS FASTER RELATIVE TO THEIR WEIGHT?

Xavier Worthy: 165 pounds, expected 4.31, ran 4.21, residual -0.10 . Jordan Davis: 341 pounds, expected 5.25, ran 4.78, residual -0.47 .



HOW CAN YOU CONVERT THE RESIDUAL TO A MORE USER FRIENDLY NUMBER?

FASTEST RAW 40S, 2019 TO 2025 COMBINES

PLAYER	POS	YEAR	WEIGHT	40	RAW SPEED GRADE	ADJ SPEED GRADE
Xavier Worthy	WR	2024	165	4.21	A+	B-
Kalon Barnes	CB	2022	183	4.23	A+	B+
DJ Turner	CB	2023	178	4.26	A+	B
Tariq Woolen	CB	2022	205	4.26	A+	A+
Henry Ruggs III	WR	2020	188	4.27	A+	B+

Raw grade: the share of every participant he outran. Adjusted grade: the share of players his weight he outran.

03

INTERACTION TERMS



**WOULD YOU EXPECT SHOT RELEASE SPEED TO MATTER MORE OR LESS
TO WIN SHARES FOR SHORTER PLAYERS?**

INTERACTION TERMS

$$Y = a + b_1X_1 + b_2X_2 + b_3X_1X_2$$

Synergistic: $b_3 > 0$. Increasing X_1 and X_2 together adds more than the sum of increasing each one by itself.

Antagonistic: $b_3 < 0$. Increasing X_1 and X_2 together adds less than the sum of increasing each one by itself.

In Excel, the interaction is one more column: X_1 times X_2 , given to LINEST with the other two.

SYNERGISTIC

$b_3 > 0$. Each one helps more when the other is higher. Five more minutes of ice time a game adds 3 goals a season for a weak shooter and 5 for a great one.

Goals = $2 + 0.5M + 1S + 0.1MS$, where M is minutes per game and S is shooting skill from 1 to 5. Each extra minute is worth $0.5 + 0.1S$ goals, so it is worth more the better the shooter. Illustrative numbers.

ANTAGONISTIC

$b_3 < 0$. Each one helps less when the other is higher. Ten more inches of vertical leap adds 5 rebounds a game for a 6-footer and only 1 for a player who is 6'10".

Rebounds = $2 + 0.5H + 0.5V - 0.04HV$, where H is inches over six feet and V is inches of vertical leap. Each extra inch of leap is worth $0.5 - 0.04H$ rebounds, so it is worth less the taller the player. Tall players already reach the ball. Illustrative numbers.

SEVERINI'S MODEL OF 40 TIME, WEIGHT AND SQUAT

Predicting 40 yard-dash time by weight and squat:

$$T = 5.0 + 0.005W - 0.02S$$

Adding in the independent variable Weight x Squat:

$$T = 5.0 + 0.005W - 0.01S + (0.00003)(W \times S)$$

As a player's weight increases, the 40-yard dash time improvement from an additional unit of strength gets smaller.

T is 40 time in seconds, W is weight and S is squat, both in pounds. Severini (2020), Analytic Methods in Sports.

04

CATEGORICAL VARIABLES

Runs allowed per 9 = $a + b_1(\text{Strikeouts per 9}) + b_2(\text{Walks per 9})$



HOW WOULD WE INCLUDE LEAGUE IN OUR RUNS ALLOWED MODEL?

DUMMY VARIABLES

TEAM RUNS ALLOWED PER NINE, 2015 TO 2019

	COEFFICIENT	STANDARD ERROR	T STAT	P-VALUE	LOWER 95%	UPPER 95%
Intercept	3.47	0.48	7.2	less than 0.0001	2.51	4.42
Strikeouts per 9	-0.214	0.041	-5.2	less than 0.0001	-0.295	-0.132
Walks per 9	0.906	0.088	10.3	less than 0.0001	0.732	1.080

Null hypothesis for each: it adds nothing once the other is in the model. 150 team-seasons, R squared 0.52.

K CATEGORIES, K – 1 DUMMIES

When there are k categories, create $k-1$ dummy variables. The category when all dummy variables are 0 is known as the "reference" category and the appropriate adjustment will be reflected in the y -intercept. Each dummy coefficient is the difference from that baseline.

A model with an intercept needs one category left out as the reference, and every dummy coefficient is read as a difference from it. In Excel, `=IF($A2=B$1, 1, 0)` creates a dummy.

CATEGORICAL VARIABLES WITH INTERACTION EFFECTS

By only bringing in categorical variables, we only adjust the y-intercept. Consider scoring as a function of offensive rebounding, where Guard is the reference category:

Full model: $Y = 10.54 + 2.977X - 6.59C - 2.73F$

Center: $Y = 3.95 + 2.977X$

Forward: $Y = 7.81 + 2.977X$

Guard: $Y = 10.54 + 2.977X$

Here we apply a consistent penalty to centers, but the reward per rebound remains the same.

X is offensive rebounds; C and F are the center and forward dummies. Severini (2020).

CATEGORICAL VARIABLES WITH INTERACTION EFFECTS

By creating additional input variables where we multiply offensive rebounds by the dummy variable, we can have different y-intercepts and slopes for each position.

Full model: $Y = 7.85 + 7.756X - 5.99C + 1.10F - 3.790XC - 5.538XF$

Center: $Y = 1.86 + 3.966X$

Forward: $Y = 8.95 + 2.218X$

Guard: $Y = 7.85 + 7.756X$

X is offensive rebounds; C and F are the center and forward dummies. Severini (2020).